



Improving I/O Performances of HPC Application Gysela

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**RESEARCH
REPORT**

N° XXXX

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Project-Teams TADaaM



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Abstract: In modern High Performances Computing's (HPC) systems, Input/Output (I/O) performances are often a limiting factor for large scale scientific simulations. Since supercomputer power and data volume keep increasing, I/O performance optimization becomes essential in order to efficiently exploit available resources. In this context, scientific applications are largely based on Parallel File System (PFS) enabling the access at the same time to very large amount of data. However, the results achieved highly depend of many configuration's parameters as well as grounds for accessing the data used by applications. Understanding and Optimizing those interactions constitutes a crucial issue for improving HPC simulations global efficiency. This work is part of the optimization of I/O performances of the application called Gysela, a simulation code developed by CEA for the study of plasma's turbulences in Tokamak, for the study of nuclear fusion. Those simulations produce large amount of data and needed regular checkpoints writing, that allow to save the state of the simulation it can resume it in case of an interruption, whether it is due to a hardware failure or a limitation on the allotted processing time on supercomputers.

Key-words: HPC, Parallel I/O, Performances Tuning, Parallel File Systems

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Mots-clés : calcul formel, base de formules, protocole, différentiation automatique, génération de code, modélisation, lien symbolique/numérique, matrice structurée, résolution de systèmes polynomiaux

1 First section

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